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tues. March 24

Boiled down fnm. 8 & 9:

$a \equiv a' \mod m$

$b \equiv b' \mod m$, then $a+b \equiv a'+b' \mod m$ &

$ab \equiv a' \cdot b' \mod m$.

Thm. 6

$a \equiv b \mod m \iff m \mid (a-b)$

$\Updownarrow$ def

$\mathrm{Rem}(a,m) = \mathrm{Rem}(b,m)$

pf Mod. Arithmetic Behaves Well Thm.

Since $a \equiv a' \mod m$, we have $m \mid (a-a')$.

So $a-a' = mk$ for some $k \in \mathbb{Z}$.

Thus $a = a' + mk$.

Likewise, $b = b' + ml$ for some $l \in \mathbb{Z}$.

We compute $a+b = (a'+mk) + (b'+ml)$.

Thus $a+b - a'+b' = m(k+l)$.

Therefore $a+b \equiv a'+b' \mod m$.

Further, $ab = (b'+ml)(a'+mk)$

$ab = a'b' + \overline{mkb' + mla' + mlmk}$ $m\mathbb{Z}$ l

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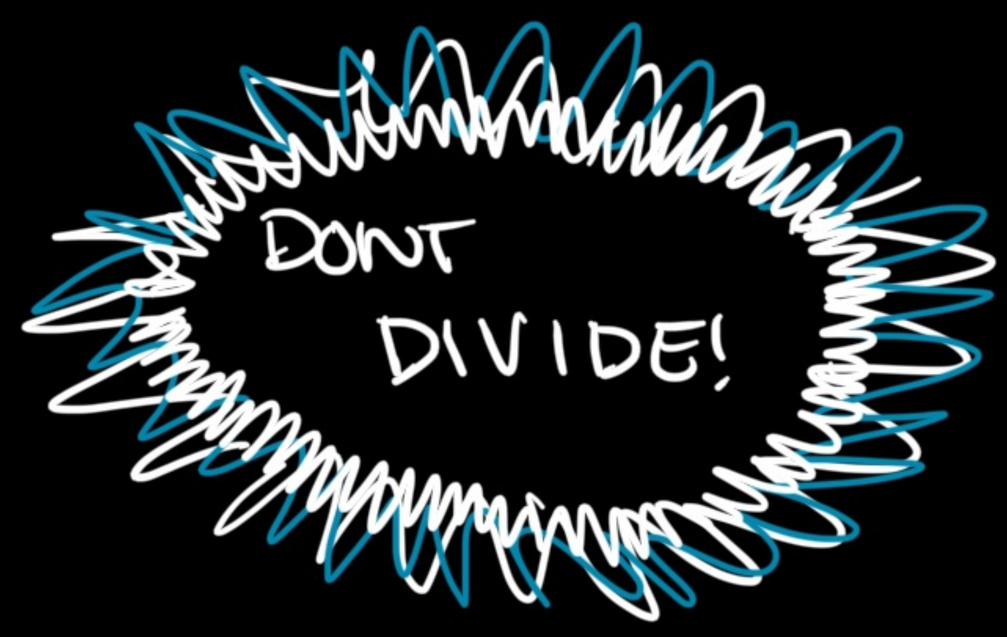
Further, $ab = (b' + ml)(a' + mk)$

$$ab = a'b' + mkb' + mla' + m^2kl$$

Thus, $ab - a'b' = m(\underbrace{kb' + la' + kl}_{\text{we know this is just an integer}})$

So $ab \equiv a'b' \mod m$. $\square$

worksheet help:



$x^2 - 1 = (x-1)(x+1)$

$x^2 - 1 \equiv (x-1)(x+1) \mod m$

$x \equiv 1 \mod m$

$x \equiv -1 \mod m$

mod 8

$\rightarrow 3^2 = 9 = 1 \mod 8$

So 3 is a solution to $x^2 - 1 \equiv 0 \mod 8$

hint for last problem

$1^2 = 1 \not\equiv 3 \mod 4$

$0^2 = 0 \not\equiv 3 \mod 4$

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hint for last problem

$1^2 = 1 \not\equiv 3 \mod 4$

$2^2 = 4 \equiv 0 \not\equiv 3 \mod 4$

$3^2 = 9 \equiv 1 \not\equiv 3 \mod 4$

$0^2 = 0 \not\equiv 3 \mod 4$

$x^2 = 3$ has no sol. mod 4

maybe b/c every #^{2} will boil down to 3, 1, or 0 which $\not\equiv$ mod 4?

Modular Arithmetic is Going Great!

But equality is so much better...

(Analogy w/ Inter-difference Equivalence $\rightsquigarrow$ Equal as integers)

If $a \equiv b \mod m$, they behave the same as far as m is concerned.

Let's lump all the <u>things that</u> behave the

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If $a \equiv b \mod m$, they behave the same as far as m is concerned.

Let's lump all the things that behave the same together.

DEF 3

Let m be a positive integer. We define equivalence classes under the equivalence relation $\equiv_m$.

congruence class
residue class

Define the equivalence class of a modulo m to be

$$[a]_m = [a] = \bar{a}_m = \bar{a} = a + m\mathbb{Z}$$

$$= \{x \in \mathbb{Z} : x \equiv a \mod m\}$$

Also

$$= \{x \in \mathbb{Z} : \mathrm{Rem}(x, m) = \mathrm{Rem}(m, x)\}$$

Also

$$= \{x \in \mathbb{Z} : m \mid (x - a)\}$$

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ex)

a. $[2]_7 = \overline{2}_7$

$= \{x \in \mathbb{Z} : x \equiv 2 \bmod 7\}$

$= \{x \in \mathbb{Z} : 7 \mid (2-x)\}$

$= \{x \in \mathbb{Z} : x = 2 + 7k \text{ for some } k \in \mathbb{Z}\}$

$= \{2 + 7k \text{ where } k \in \mathbb{Z}\}$

$= \{\ldots, -19, -12, -5, 2, 9, 16, 23, 30 \ldots\}$

b. $[a]_m = \{a + mk : k \in \mathbb{Z}\}$

* this is a set of integers w/infinetly many elements

$[2]_7 \equiv \cancel{9 \bmod 7}$ don't do this

$[2]_7 = [9]_7$ yay use this!

→ Let $a \in [2]_7$. By def $a \equiv 2 \bmod 7$, since

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We denote the set of all equivalence classes mod m by $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z} = \{[a]_m : a \in \mathbb{Z}\}$.

We call this the set of integers modulo m or $\mathbb{Z}$ mod m or $\mathbb{Z}$ mod $m\mathbb{Z}$

ex)

a. $\mathbb{Z}/3\mathbb{Z} = \mathbb{Z}_3 = \{[a]_3 : a \in \mathbb{Z}\}$

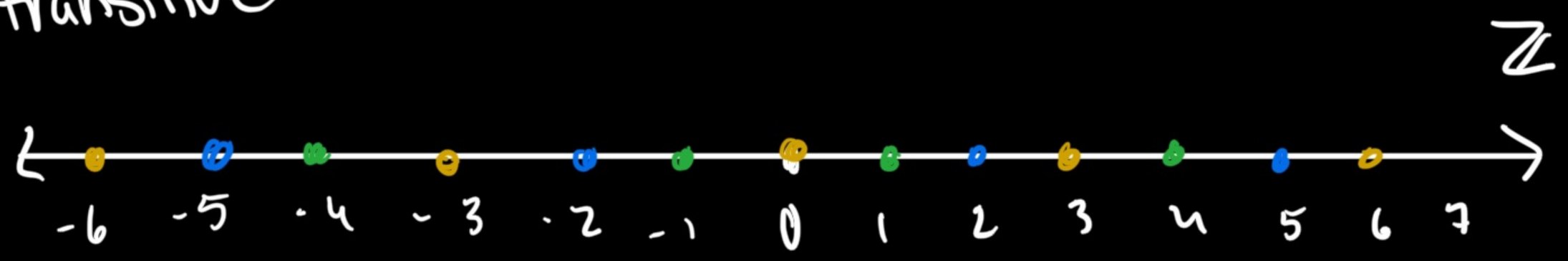
$= \{[0]_3, [1]_3, [2]_3\}$

b. $[7]_3 = \{x \in \mathbb{Z} : x \equiv 7 \bmod 3\}$

$= \{x \in \mathbb{Z} : x \equiv 1 \bmod 3\}$

$7 \equiv 1 \bmod 3$

$\equiv_3$ is transitive



$[0]_3$ $[1]_3$ $[2]_3$

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$\boxed{\text{Thm. 12}}$

$a \equiv b \mod m \iff [a]_m = [b]_m$

<u>Pf</u>

$\Rightarrow$

If $[a]_m = [b]_m$, note $a \equiv a \mod m$, so $a \in [a]_m$. Thus $a \in [b]_m$ b/c they're $=$ so $a \equiv b \mod m$.

$\Leftarrow$

Let $x \in [a]_m$, thus $x \equiv a \mod m$, & $a \equiv b \mod m$ by the transitivity of congruence modulo m, we see $x \equiv b \mod m$. Thus $x \in [b]_m$, and $[a]_m \subseteq [b]_m$. Conversly, Let $y \in [b]_m$, thus $y \equiv b \mod m$, & $b \equiv a \mod m$ by the transitivity of congruence modulo m, we see $y \equiv a \mod m$. $\square$

other direction

"a symmetric argument holds"

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Corollary 13

$[a]_m = [b]_m \iff a \equiv b \mod m \iff \mathrm{Rem}(a,m) = \mathrm{Rem}(b,m)$

$\iff m \mid (a-b)$

pf

Combine thm. 12, thm. 6, & def. of $\equiv_m$.

$\square$

Thm. 14

Let $[a] \in \mathbb{Z}/m\mathbb{Z}$ be a congruence class of a. Then $\exists$ exactly one $b \in \{0,1,\dots,m-1\}$ s.t. $[a]_m = [b]_m$.

pf

Corollary 5 + Thm. 12. $\square$

Corollary 15

The rule $x \mapsto [x]_m$ defines a bijection between $\{0,\dots,m-1\}$ & $\mathbb{Z}/m\mathbb{Z}$.

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"be a congruence class of"

Thm. 14

Let $[a] \in \mathbb{Z}/m\mathbb{Z}$. Then $\exists$ exactly one $b \in \{0, 1, \dots, m-1\}$ s.t. $[a]_m = [b]_m$.

<u>pf</u>

Corollary 5 + Thm. 12. $\square$

<u>Corollary 15</u>

The rule $x \mapsto [x]_m$ defines a bijection between $\{0, \dots, m-1\}$ & $\mathbb{Z}/m\mathbb{Z}$.

<u>Cor. 16</u>

$\mathbb{Z}/m\mathbb{Z}$ has m elements. $|\mathbb{Z}/m\mathbb{Z}| = m$

In particular, $\mathbb{Z}/m\mathbb{Z} = \boxed{\{}[0], [1], \dots, [m-1]\}$

$\Downarrow$

$[-1]$